

NovaPharma Iberia — Step 4.C Review: Visuals

Digital transformation project · Excel → MySQL / Python / Power BI · Author: Nix · 25 September 2026

Starting point

The model (4.A) and the measures (4.B) were ready, and the pages had been decided in 4.0. What was left was presentation: layout, colours and how the report behaves when someone clicks.

Tools

The report was saved as a Power BI project (PBIP), which stores every page and visual as a text file. That made it possible to build the pages with an open-source MCP server (powerbi-report-mcp, MIT). It places visuals, applies themes, sets sorting and interactions, and never touches the model. Its code was reviewed and the plugin was compiled from source before using it. The model and the DAX stayed manual on purpose.

One theme for everything

Instead of formatting visual by visual, a custom theme ("NovaPharma Clinical") sets the look once: light grey canvas, white cards with rounded corners and a soft shadow, and a navy header with a thin teal line. Actual is always blue and budget always grey. Only one colour carries meaning: **green or red for Var vs Budget %**, in the cards and in the tables. If everything is coloured, nothing stands out.

The pages

Page	What it shows
Overview	4 KPI cards · Net sales vs budget by month · Performance by region · Net sales by brand (drill to SKU)
Budget vs Actual	Actual, budget and variance · Actual vs budget by month · Variance by brand, from worst to best
About	What the report answers, how the data gets here, definitions, the project

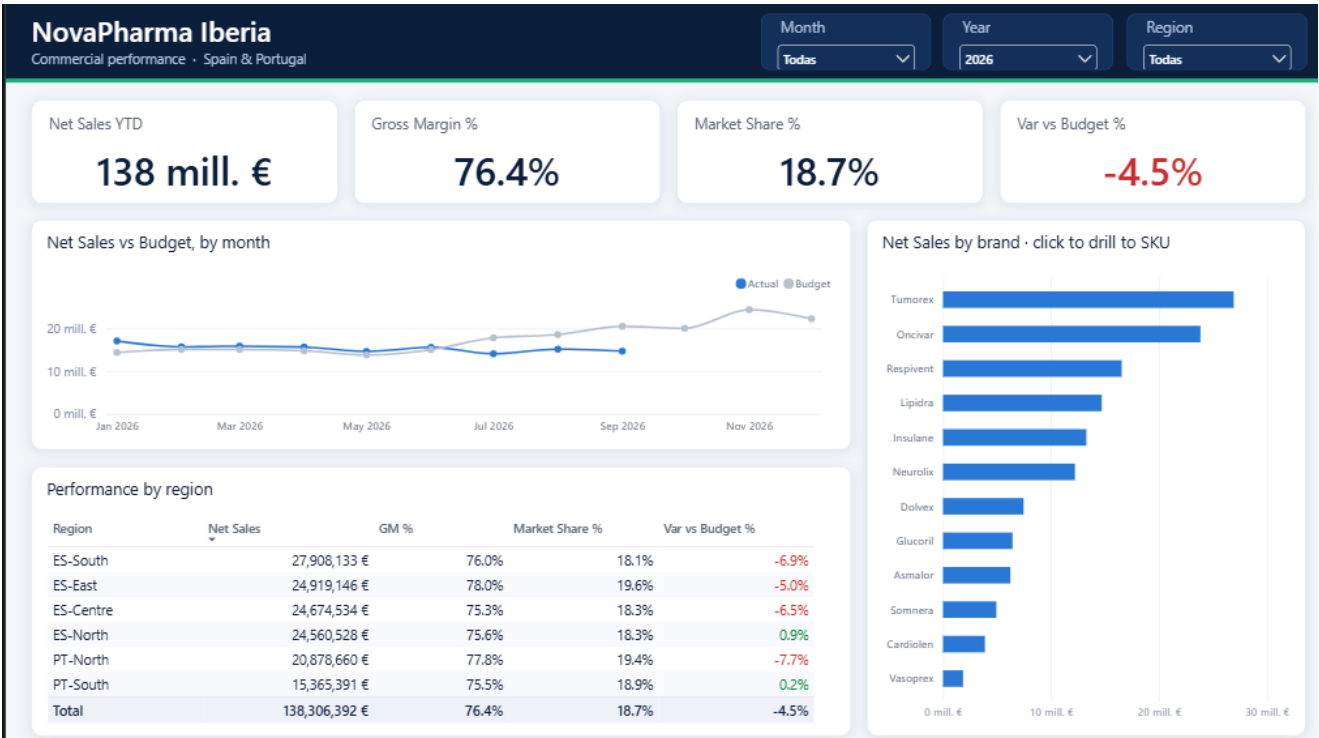
The header is the same on every page, with three slicers: **Month, Year and Region**. Month replaces the Excel's CurrentMonth cell: pick a month and the YTD card stops there, while the monthly charts keep the full year for context. There is no channel slicer on purpose. Channel only exists on the sales side, so it would distort the budget and market figures.

Iterating

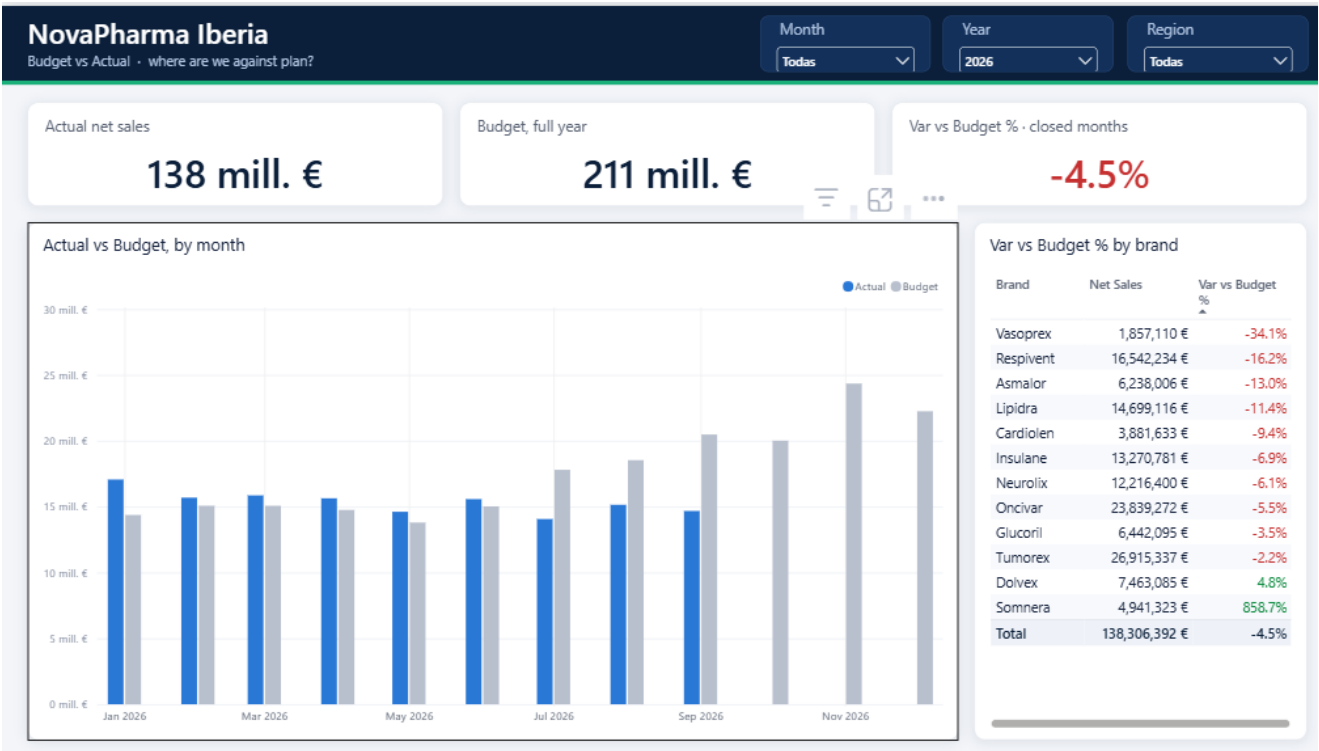
The first draft looked fine as files and not so fine on screen: tiny card values, a cut header and an outlier crushing a chart. Three rounds of fix → reopen → screenshot solved it. A few decisions came out of that loop. The brand variance became a sorted table, because a launch brand at +858 % flattened the bar chart. The therapeutic-area chart was removed to give brands the full column. The monthly axis now starts at zero.

Result: 3 pages, 1 theme, and every number traceable to a validated measure. The Excel's 17 sheets and its dashboard are now a report that updates by itself every month.

Screenshots



Page 1 — Overview (Year 2026, all regions)



Page 2 — Budget vs Actual

What this report answers

How are we doing this year? Net sales YTD, gross margin, market share and the gap to budget.

How is it trending? Actual vs budget, month by month.

Where does it come from? By brand (drill to SKU) and by region.

Pages: Overview · Budget vs Actual · About.

Definitions

Net Sales = gross sales – discounts – rebates (calculated in SQL).

Gross Margin % = gross margin ÷ net sales of the selection.

Net Sales YTD = January to the latest month selected.

Market Share % = net sales ÷ IQVIA market value (brand level).

Var vs Budget % = (actual – budget) ÷ budget, closed months only.

Data is synthetic: company, brands and figures are fictional.

How the data gets here

1. ERP monthly extract (CSV): the file the Excel model pasted by hand.
2. Python ETL: validates every row, quarantines bad ones, loads only new data. Runs by itself every month.
3. MySQL warehouse novapharma_dw: star schema, 11 tables, 2 SQL views.
4. Power BI: Import mode, 11 relationships, 9 DAX measures.

The project

From Excel to Data Warehouse: a digital transformation case.

A 17-sheet Excel commercial model, replaced by a SQL data model, a Python pipeline, a MySQL warehouse and this Power BI report.

Built by Nix · 2026

github.com/nixxx10/novapharma-excel-to-datawarehouse